

Your Thesis Title Here

Optional Subtitle

Your Full Name

A thesis submitted in partial fulfilment of the requirements for the degree of
Master of XXX in XXX,
The University of Auckland, [year of completion].

Abstract

Your abstract here.

Keywords: keyword1, keyword2, keyword3

Dedication

To my family and mentors.

Acknowledgements

I would like to express my sincere gratitude to all those who supported me throughout this research journey.

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Glossary

Term	Definition
$\text{\LaTeX}$	Typesetting system commonly used in academic and scientific documents

Acronyms

Abbreviation	Full Form
UoA	University of Auckland
MSD	Mean Square Displacement
RDF	Radial Distribution Function
MD	Molecular Dynamics

Nomenclature

Symbol	Description	Unit
v	Velocity	m/s
x	Position	m
c	Speed of light in vacuum	3.00×10^8 m/s
h	Planck constant	6.63×10^{-34} J·s

Chapter 1

Introduction

1.1 Background

Your introduction text here.

1.2 Research Aims and Objectives

State your research aims and objectives.

1.3 Thesis Structure

Outline the structure of this thesis.

Chapter 2

Literature Review

2.1 Theoretical Framework

Review of theoretical foundations.

2.2 Empirical Studies

Review of relevant empirical research.

2.3 Research Gap

Identify the gap this thesis addresses.

Chapter 3

Methodology

3.1 Research Design

Describe your overall research approach.

3.2 Data Collection

Describe data collection procedures.

3.3 Data Analysis

Describe analytical methods.

3.4 Ethical Considerations

Address ethical aspects of the research.

Chapter 4

Results

4.1 Overview

Present your findings.

4.2 Detailed Analysis

Present detailed results with tables and figures.

Chapter 5

Discussion

5.1 Interpretation of Findings

Discuss findings in relation to the literature.

5.2 Implications

Theoretical and practical implications.

5.3 Limitations

Acknowledge study limitations.

Chapter 6

Conclusion

6.1 Summary of Findings

Summarise key findings.

6.2 Recommendations

Provide recommendations for future research or practice.

Chapter 7

Template Feature Demonstrations

This chapter demonstrates all the common features available in this template. Use it as a reference when writing your thesis.

7.1 Text Formatting

Bold text and *italic text* and SMALL CAPS. You can also combine them: ***bold italic***.

7.2 Citations and Paragraph Formatting

7.2.1 In-Text Citation Examples

The following examples demonstrate common citation patterns using the APA style.

Parenthetical citations — the most common form: several studies have investigated deep learning for image recognition (Chen et al., 2023). Convolutional neural networks remain the dominant architecture in this domain (Liu & Patel, 2024), though recent work suggests alternatives (Author, 2024).

Narrative citations — the author's name appears as part of the sentence: Scholar (2023) provided a comprehensive overview of academic writing conventions. More recently, Chen et al. (2023) surveyed over 200 papers on natural language processing applications. In her doctoral work, Kim (2023) developed novel computational methods for genomic analysis.

Multiple citations together: prior work has addressed this problem from several angles (Author, 2024; Chen et al., 2023; Liu & Patel, 2024). When citing multiple sources, APA style orders them alphabetically and separates with semicolons.

Citing specific parts: gradient-based methods are widely adopted (see Patel, 2024, chap. 3), and the implementation details are well-documented (Moore, 2024). Technical standards also provide guidance on data formats (Group, 2024).

Year-only and page citations: the foundational work appeared in 2023, and later extensions can be found in Wang (2024, p. 45).

7.2.2 Paragraph Spacing and Indentation

The following two paragraphs demonstrate the default paragraph formatting in this template. Paragraphs are separated by vertical space (1.5-line spacing throughout) with no first-line indentation, as commonly used in theses following UoA guidelines.

The advent of deep learning has fundamentally transformed the landscape of artificial intelligence research over the past decade. Convolutional neural networks (Liu & Patel, 2024), recurrent architectures, and more recently transformer-based models have achieved state-of-the-art performance across a wide range of tasks including computer vision, natural language processing, and speech recognition. This rapid progress has been driven by the convergence of several key factors: the availability of large-scale annotated datasets, the development of powerful graphics processing units for parallel computation, and algorithmic innovations in training deep networks. The impact of these advances extends beyond academic research into industrial applications, where deep learning systems now power everything from recommendation engines to autonomous vehicles.

Despite these remarkable achievements, significant challenges remain. Modern deep learning models require enormous computational resources for training, raising concerns about energy consumption and environmental impact (Chen et al., 2023). Moreover, these models often function as black boxes, making it difficult to interpret their decisions or to guarantee their reliability in safety-critical applications. Researchers have proposed various approaches to address these limitations, including model compression techniques (Wang, 2024), interpretability methods, and the integration of domain knowledge through hybrid architectures (Patel, 2024). Addressing these challenges is essential for the responsible deployment of AI systems in healthcare, transportation, and other domains where errors can have serious consequences.

7.3 Lists

7.3.1 Unordered List (itemize)

The following factors were considered:

- Data quality and availability
- Computational resources
- Prior work in the field
 - Sub-factor A — nested lists are supported
 - Sub-factor B
- Ethical considerations

7.3.2 Ordered List (enumerate)

The experiment proceeded as follows:

1. Collect raw data from the sensors.
2. Pre-process the data using standard normalisation.
3. Train the model for 100 epochs.
4. Evaluate on the held-out test set.

7.3.3 Roman Numeral List

- i. First criterion: the algorithm must converge within 1000 iterations.
- ii. Second criterion: the error rate must be below 5%.
- iii. Third criterion: the model must generalise to unseen data.

7.3.4 Numbered List with Parentheses

- 1) First step: collect raw data from the sensors.
- 2) Second step: pre-process the data using standard normalisation.
- 3) Third step: train the model and evaluate performance.

7.4 Mathematical Equations

7.4.1 Inline Equation

The well-known mass–energy equivalence is $E = mc^2$, where m is the rest mass and c is the speed of light in vacuum.

7.4.2 Display Equation (Unnumbered)

The gravitational force between two masses is given by:

$$F = G \frac{m_1 m_2}{r^2}$$

where G is the gravitational constant.

7.4.3 Numbered Equation (Cross-Referenced)

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i; \theta))^2 \quad (7.1)$$

Equation 7.1 defines the mean-squared-error loss function commonly used in regression tasks.

7.4.4 Multi-Line Aligned Equations

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \quad (7.2)$$

$$\hat{\beta} = (X^\top X)^{-1} X^\top y \quad (7.3)$$

Equations 7.2 and 7.3 show the linear regression model and its ordinary-least-squares solution.

7.4.5 Case-Based Definition

$$\text{ReLU}(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (7.4)$$

Equation 7.4 defines the Rectified Linear Unit activation function.

7.5 Algorithms / Pseudocode

Algorithm 1 demonstrates a simple gradient descent optimisation procedure using the `algorithm2e` package.

Algorithm 1: Gradient Descent Optimisation

Parameters: Learning rate η , max iter. T

Initialise: Model parameters θ_0

```
1 for  $t \leftarrow 1$  to  $T$  do
2   | Compute gradient  $\nabla \mathcal{L}(\theta_{t-1})$ ;
3   | Update  $\theta_t \leftarrow \theta_{t-1} - \eta \nabla \mathcal{L}(\theta_{t-1})$ ;
4   | if  $\|\nabla \mathcal{L}(\theta_t)\| < \epsilon$  then
5   |   | break;
6 return  $\theta_t$ ;
```

Line 3 of Algorithm 1 performs the standard parameter update step.

7.6 Figures

7.6.1 Single Figure

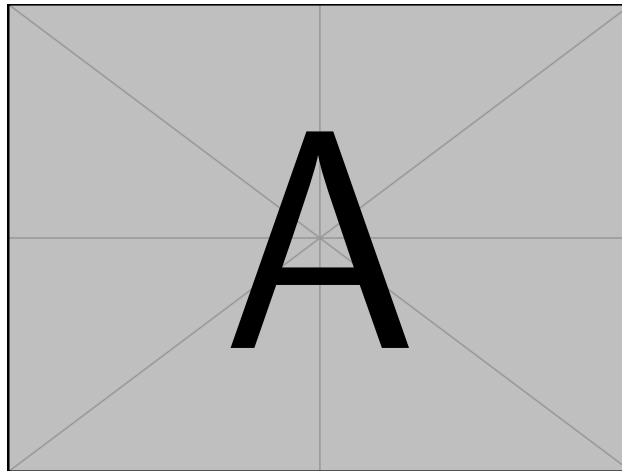


Figure 7.1 An example figure showing the experimental apparatus

7.6.2 Side-by-Side Figures (Subfigures)

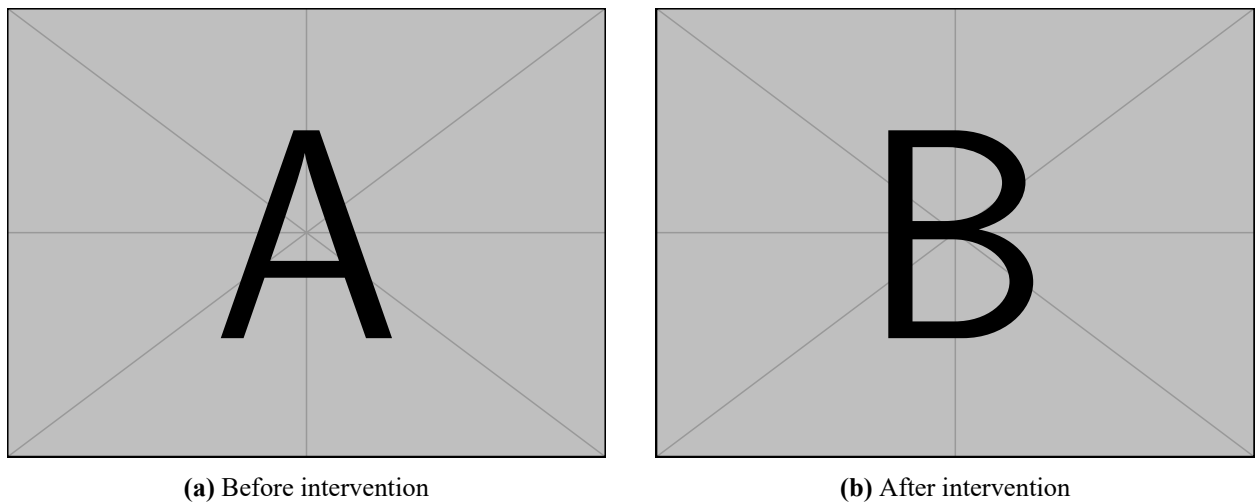


Figure 7.2 Comparison of results before and after the intervention.



Figure 7.3 UoA logo as an example of including a figure from the template's own directory

Figure 7.1 shows the setup, and Figure 7.2 compares the two conditions.

7.7 Tables

7.7.1 Standard Three-Line Table

Table 7.1 Performance comparison of three algorithms on the benchmark dataset

Algorithm	Accuracy (%)	Precision	Recall
Baseline	87.3	0.85	0.86
Method A	94.2	0.93	0.92
Method B	96.7	0.96	0.95
Method C	97.1	0.97	0.96

Note. Higher values indicate better performance.

7.7.2 Table with Partial Rules (cmidrule)

Table 7.2 Summary of participant demographics across two cohorts

Metric	Cohort A		Cohort B	
	Mean	SD	Mean	SD
Age (years)	24.3	3.1	31.7	4.5
Score	78.5	9.2	82.1	7.8

As shown in Table 7.1, Method C achieves the best performance. Table 7.2 summarises the participant information.

7.7.3 Rotated Full-Page Table (Landscape)

For wide tables that cannot fit on a portrait page, use the `landscape` environment (from `pdf1scape` package) to rotate the page 90°. The PDF viewer will automatically show the page in landscape orientation.

Table 7.3 Comprehensive comparison of model performance across twelve datasets and six methods. This table is intentionally wide to demonstrate the landscape (rotated) page layout.

Dataset	Accuracy (%)						Precision						Recall					
	A	B	C	D	E	F	A	B	C	D	E	F	A	B	C	D	E	F
CIFAR-10	94.2	95.1	95.8	96.3	96.8	97.3	0.94	0.95	0.95	0.96	0.96	0.97	0.93	0.94	0.95	0.96	0.96	0.97
ImageNet	76.5	77.2	78.0	79.5	80.4	82.1	0.75	0.76	0.77	0.79	0.80	0.81	0.74	0.75	0.77	0.78	0.79	0.80
MNIST	99.1	99.2	99.3	99.4	99.5	99.6	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
SVHN	96.3	96.5	96.8	97.0	97.5	98.0	0.96	0.96	0.97	0.97	0.97	0.98	0.95	0.96	0.96	0.97	0.97	0.98
COCO	72.4	73.1	74.5	75.8	76.8	78.5	0.71	0.72	0.73	0.75	0.76	0.78	0.70	0.71	0.73	0.74	0.75	0.77
Pascal VOC	85.2	86.0	87.3	89.0	90.2	91.4	0.84	0.85	0.86	0.88	0.89	0.91	0.83	0.84	0.85	0.87	0.88	0.90
Cityscapes	80.1	81.4	82.5	83.8	84.2	86.7	0.79	0.80	0.82	0.83	0.83	0.86	0.78	0.79	0.81	0.82	0.82	0.85
ADE20K	45.3	47.1	48.7	50.2	52.0	55.6	0.44	0.45	0.47	0.50	0.51	0.55	0.43	0.44	0.46	0.49	0.50	0.54
KITTI	88.2	89.5	90.3	91.7	92.4	93.1	0.87	0.88	0.90	0.91	0.92	0.93	0.86	0.87	0.89	0.90	0.91	0.92
SUN RGB-D	62.1	64.3	66.8	68.5	71.2	73.9	0.61	0.63	0.66	0.68	0.70	0.73	0.60	0.62	0.65	0.67	0.69	0.72
NYU Depth	58.7	60.5	63.2	65.8	68.1	70.4	0.57	0.59	0.62	0.65	0.67	0.70	0.56	0.58	0.61	0.64	0.66	0.69
ScanNet	71.4	73.2	75.6	77.9	80.3	82.7	0.70	0.72	0.74	0.77	0.79	0.82	0.69	0.71	0.73	0.76	0.78	0.81

Note. A = Baseline; B = ResNet-50; C = ResNet-101; D = ViT-B/16; E = Swin-T; F = Ours. Best per dataset in each metric is underlined.

7.8 Long Quotation (UoA-Specific)

This environment follows the University of Auckland guidelines exactly. According to Smith (Author, 2024):

This is a demonstration of a long quotation spanning three or more lines. Per the University of Auckland formatting guidelines, it is indented 10 mm from the left-hand margin, single-spaced, and has no quotation marks. The text appears in a slightly smaller font to visually distinguish it from the surrounding body text. This passage continues to demonstrate the required formatting for long quotations in sub-doctoral theses at the University of Auckland.

The findings above have significant implications for future research in this area. As highlighted by the quotation, the formatting aligns precisely with institutional requirements.

7.9 Footnotes

This statement is supported by empirical evidence¹. Additional findings corroborate this conclusion².

¹See the comprehensive review by Author (2024) for a detailed discussion of the underlying methodology.

²Recent meta-analyses confirm these results across multiple independent studies (Scholar, 2023).

Appendix A

Additional Data

Your appendix content here. Each appendix should be labelled in sequence with capital letters or numerals.

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